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The error ΔZ in Z is then given by

$$Z \pm \Delta Z = (A \pm \Delta A) + (B \pm \Delta B)$$
$$= (A + B) \pm (\Delta A + \Delta B)$$
$$= Z \pm (\Delta A + \Delta B)$$
$$\text{(or)} \quad \Delta Z = \Delta A + \Delta B \qquad (1.3)$$

The maximum possible error in the sum of two quantities is equal to the sum of the absolute errors in the individual quantities.

EXAMPLE 1.5

Two resistances $R_1 = (100 \pm 3)\ \Omega$, $R_2 = (150 \pm 2)\ \Omega$, are connected in series. What is their equivalent resistance?

Solution

$$R_1 = 100 \pm 3\Omega;\ R_2 = 150 \pm 2\Omega$$

Equivalent resistance R = ?

Equivalent resistance $R = R_1 + R_2$

$$= (100 \pm 3) + (150 \pm 2)$$
$$= (100 + 150) \pm (3 + 2)$$
$$R = (250 \pm 5)\ \Omega$$

(ii) Error in the difference of two quantities

Let ΔA and ΔB be the absolute errors in the two quantities, A and B, respectively. Then,

Measured value of $A = A \pm \Delta A$

Measured value of $B = B \pm \Delta B$

Consider the difference, $Z = A - B$

The error ΔZ in Z is then given by

$Z \pm \Delta Z = (A \pm \Delta A) - (B \pm \Delta B)$

$$= (A - B) \pm \Delta A \mp \Delta B$$
$$= Z \pm \Delta A \mp \Delta B$$
$$\text{(or)} \quad \Delta Z = \Delta A + \Delta B \ldots\ldots\ldots\ldots\ldots\ldots (1.4)$$

The maximum error in difference of two quantities is equal to the sum of the absolute errors in the individual quantities.

EXAMPLE 1.6

The temperatures of two bodies measured by a thermometer are $t_1 = (20 + 0.5)°C$, $t_2 = (50 \pm 0.5)°C$. Calculate the temperature difference and the error therein.

Solution

$t_1 = (20 \pm 0.5)°C \quad t_2 = (50 \pm 0.5)°C$

temperature difference t=?

$t = t_2 - t_1 = (50 \pm 0.5) - (20 \pm 0.5)°C$

(Using equation1.4)

$= (50 - 20) \pm (0.5 + 0.5)$

$t = (30 \pm 1)°C$

(iii) Error in the product of two quantities

Let ΔA and ΔB be the absolute errors in the two quantities A, and B, respectively. Consider the product $Z = AB$

The error ΔZ in Z is given by $Z \pm \Delta Z = (A \pm \Delta A)\ (B \pm \Delta B)$

$$= (AB) \pm (A\ \Delta B) \pm (B\ \Delta A) \pm (\Delta A\ .\ \Delta B)$$

Dividing L.H.S by Z and R.H.S by AB, we get,

$$1 \pm \frac{\Delta Z}{Z} = 1 \pm \frac{\Delta B}{B} \pm \frac{\Delta A}{A} \pm \frac{\Delta A}{A} \cdot \frac{\Delta B}{B}$$